AI-Based Electrical Insulation Degradation Classification Using Leakage Current Patterns

INTERNSHIP CAPSTONE PROJECT REPORT

Abstract

Electrical insulation degradation is an important reliability concern in electrical equipment. Leakage current contains useful information about insulation condition. This project develops an artificial-intelligence-based classification system that uses leakage-current patterns to classify insulation into Healthy, Moderately Degraded, and Severely Degraded conditions. Signal-processing features such as RMS value, peak current, standard deviation, crest factor and frequency-domain information are extracted from leakage-current samples. Random Forest and Support Vector Machine (SVM) classifiers are trained and evaluated using labelled data. A simple Streamlit dashboard is proposed for entering or uploading leakage-current data and displaying the predicted insulation condition. For demonstration, the project can use simulated labelled signals when a real experimental dataset is unavailable; such results must be clearly identified as simulated validation rather than field performance.

1. Introduction

Electrical insulation prevents unwanted current flow between conductors and from conductors to earth. With ageing, thermal stress, moisture, contamination, electrical stress and mechanical effects, insulation performance can deteriorate. Leakage current is one measurable indicator that can change as insulation condition changes. Manual interpretation of large signal records can be difficult, so machine learning can help identify patterns consistently.

2. Problem Statement

Develop a software-based classification system that receives leakage-current patterns, extracts meaningful electrical and statistical features, and classifies the insulation condition into three categories: Healthy, Moderately Degraded, or Severely Degraded.

3. Objectives

1. Generate or acquire labelled leakage-current signals. 2. Preprocess the signal data. 3. Extract time-domain and frequency-domain features. 4. Train Random Forest and SVM classifiers. 5. Evaluate the models using accuracy, precision, recall and F1-score. 6. Visualize leakage-current waveforms and spectra. 7. Provide a simple Streamlit interface for prediction.

4. System Architecture

Input Leakage Current Signal → Preprocessing → Feature Extraction → Feature Scaling/Selection → Machine Learning Model → Insulation Condition Classification → Dashboard/Report.

5. Methodology

The signal is first cleaned and divided into suitable samples or windows. Time-domain features include RMS, peak magnitude, mean, standard deviation and crest factor. Frequency-domain characteristics can be obtained using FFT. The labelled feature matrix is divided into training and test sets. Random Forest provides an ensemble decision model, while SVM finds a separating boundary in feature space. The predicted class is compared with the known test label.

6. Feature Extraction

RMS = sqrt((1/N) Σ x_i^2). Peak current = max(|x_i|). Standard deviation describes signal variation. Crest factor = Peak/RMS. FFT magnitude provides frequency-domain information and can help distinguish different leakage-current patterns. These features convert a raw waveform into a compact numerical representation suitable for machine learning.

7. Machine Learning Models

Random Forest combines multiple decision trees and is useful for nonlinear relationships and mixed feature behaviour. Support Vector Machine (SVM) constructs a decision boundary that maximizes class separation; an RBF kernel can model nonlinear patterns. Both models are trained on the same labelled feature set for comparison.

8. Classification Classes

Healthy: comparatively stable and low-variation leakage-current pattern. Moderately Degraded: noticeable increase or variation in leakage current. Severely Degraded: substantially abnormal or highly variable pattern. Thresholds and class definitions must be established from validated experimental data before using the system for real equipment decisions.

9. Evaluation

The project should report a confusion matrix and class-wise precision, recall and F1-score, along with overall accuracy. Accuracy alone is not sufficient if the classes are imbalanced. The final report should distinguish simulated-data performance from performance obtained using real measured data.

10. Dashboard

A Streamlit dashboard can provide file upload or sample-data input, waveform visualization, feature values, model selection, and the predicted insulation class. The interface should clearly show that the prediction is an engineering decision-support result and not a replacement for electrical safety procedures or insulation testing standards.

11. Advantages

• Automated pattern classification. • Fast analysis of recorded signals. • Repeatable feature-based evaluation. • Supports visualization and comparison of models. • Can be extended with real laboratory measurements.

12. Limitations

Model quality depends strongly on the quality and representativeness of labelled data. Simulated signals may not capture all real insulation phenomena. Sensor noise, temperature, humidity, equipment type and measurement setup can affect leakage current. Therefore, real-world deployment requires laboratory/field validation and appropriate electrical safety practices.

13. Future Scope

Future work can include real-time sensor acquisition, larger experimental datasets, deep-learning models such as 1-D CNNs, anomaly detection, edge deployment, trend monitoring, explainable AI, and integration with condition-monitoring systems.

14. Conclusion

The proposed system demonstrates how leakage-current signal features and machine learning can be combined to classify electrical insulation condition. Random Forest and SVM provide practical baseline models, while waveform and FFT visualization improve interpretability. With properly labelled experimental data and validation, the approach can become a useful condition-monitoring aid.

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